

EXECUTIVE CV

Mohammed Elghanam

R&D CTO | Solution Architect | Engineering Consultant

Embedded & IoT Product Development | Smart Metering, LPWAN, FPGA & Safety-Critical Systems



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Featured-ready CV for senior R&D leadership, consulting, partnerships, and global technology collaboration

Executive Profile

R&D CTO and Engineering Consultant with 25+ years of experience leading embedded systems, IoT product development, smart utilities, industrial automation, FPGA-based systems, hardware/firmware integration, and safety-critical technology development. Proven track record across utilities, industrial control, telecommunications, smart infrastructure, medical device R&D, and mission-critical communication platforms.

25+

Years R&D Leadership

5

Granted Patents

GCC+

Egypt | GCC | Remote

R&D

CTO / Consulting



Strategic Value Proposition

Transforms technical ideas into practical, scalable, and market-aligned engineering solutions by combining technology strategy, system architecture, embedded hardware, firmware, FPGA, validation, and cross-functional leadership.



Target Senior Roles

R&D CTO | Solution Architect | Engineering Consultant |
Technical Advisor | Embedded & IoT Product
Development Leader | Hardware Product Development
Lead

Core Expertise

Embedded Systems Architecture

IoT & LPWAN Product Development

Smart Metering & AMI

FPGA-Based System Design

Hardware/Firmware Integration

RTU & Industrial Automation

Power Grid Protection

Safety-Critical Systems

Medical Device R&D

Technology Strategy & R&D Leadership

Apr 2026 - Jun 2026

EXPERIENCE

Solution Architect
Sanabil Solutions - "R&D as a Service"

Supported technology solution design, technical planning, and R&D-oriented product-development activities. Applied solution architecture, embedded systems, IoT, and hardware/software integration thinking to support technical decision-making.

2018 - Oct 2025

EXPERIENCE

R&D CTO
Engineering Office for Integrated Projects (EOIP) / DLECS Lab R&D

Led R&D strategy, technology direction, system architecture, and advanced engineering development across embedded systems, IoT platforms, smart utilities, FPGA-based systems, industrial automation, medical device R&D, and safety-critical technology solutions.

Aug 2000 - Dec 2017

EXPERIENCE

R&D / Embedded Systems Engineering Leader
Engineering Office for Integrated Projects (EOIP)

Led and contributed to embedded systems, industrial electronics, hardware/firmware integration, communication systems, signal processing, FPGA-based platforms, and product-development projects across utility, automation, telecom, and safety-oriented applications.

Feb 1998 - Jul 2000

EXPERIENCE

Technical Engineer / Reserve Officer
Egyptian Armed Air Force

Supported analysis and troubleshooting of control and power-system faults related to helicopter systems, gaining practical exposure to mission-critical engineering environments.

Sep 1998 - Aug 1999

EXPERIENCE

Teaching & Research Assistant
Faculty of Engineering, Banha University

Supported engineering education in the Automatic Control Department, including electrical circuit design, microelectronics circuit design, programming, and microprocessor fundamentals.

Experience Keywords

R&D Leadership | Solution Architecture | Embedded Systems | IoT Product Development | Smart Utilities | FPGA | RTU | Industrial Automation | Medical Device R&D | Safety-Critical Systems

A curated project portfolio highlighting engineering challenges, solutions, technical role, and technology stack across embedded systems, IoT, utilities, industrial automation, power systems, medical control, and mission-critical platforms.

01

Smart Utilities & Smart Metering Solutions

Led smart and prepaid metering platforms for electricity, water, and gas with embedded hardware, AMI concepts, LPWAN, RS485, cellular, Wi-Fi, PLC, data platforms, and billing integration.

02

Patented LPWAN IoT Platform

Spearheaded an end-to-end LPWAN IoT platform covering field devices, gateways, embedded software, cloud connectivity, and mobile applications for smart lighting, waste, agriculture, and buildings.

03

Advanced RTU / Embedded EMS Platforms

Led advanced RTU and embedded EMS platforms for industrial monitoring and control, including architecture definition, hardware/firmware integration, industrial communication, validation, and cybersecurity considerations.

04

Power Grid Protection & Control Systems

Directed digital overcurrent and earth-fault relay work for 11kV networks, including hardware/software development, protection testing, commissioning, and Foxboro DCS integration.

05

Mission-Critical Communication Platforms

Contributed to secure, high-performance communication platforms using FPGA-based implementation, high-speed processing, MMI integration, hardware review, and system validation.

06

Prototype SDR Platform Hardware Review

Provided hardware design review for SDR prototype boards including ARM processor, Xilinx FPGA, DDR3 RAM, RF front-end board review, signal integrity, and EMC guidelines.

07

Medical Device R&D Systems

Led ventilator and oxygen concentrator control-system R&D. Systems received calibration and safety certification from Cairo University Medical Calibration and Safety Center; Egyptian Drug Authority licensing procedures are in progress.

08

Unified Azan & Railway Safety Systems

Architected Unified Azan synchronization using GPS timing, RDS, high-power amplifiers, and networks. Contributed to railway level-crossing control systems with signal processing and safety interlocks.

Innovation & Recognition

5 granted patents related to EOIP innovation and international recognition at IIFME Kuwait for the invention "Upgrading of Traditional Meters into Smart Meters".

Core Technical Skills

Embedded & IoT

Embedded Systems, IoT Product Development, LPWAN, Smart Utilities, AMI Solutions, Prepaid Metering, RTU, Embedded EMS

Hardware / FPGA

Hardware Design, PCB Design, High-Speed Digital Design, Signal Integrity, EMC Considerations, FPGA, VHDL, Digital Design

Industrial / Power

Industrial Automation, Control Systems, Power Grid Protection, Digital Relay Systems, Overcurrent Protection, Earth Fault Protection

Leadership

R&D Leadership, Solution Architecture, Technology Strategy, Product Development, Technical Advisory, Engineering Consulting, Project Management

Education

B.Sc. Automatic Control & Instrumentation Engineering
Faculty of Engineering, Banha University | 1992 - 1997
Grade: Very Good with Honors
Graduation Project: Current & Speed PID Control of DC Separately Excited Motor

Certifications & Professional Training

- Foundations of CENELEC - TÜV SÜD Rail GmbH, Braunschweig, Germany | Feb 2015
- PMP Exam Preparation Course | Mar-Apr 2013
- VHDL Coding Training - Mentor Graphics / Cairo Faculty of Engineering, Communication Department
- PCB Design According to Digital Constraints - Mentor Graphics Egypt
- PCB Printing & Component Insertion Training - KATRON

Professional Credibility & Accuracy Notes

- 5 granted patents related to EOIP embedded/IoT and smart metering innovation.
- IIFME Kuwait recognition for the invention "Upgrading of Traditional Meters into Smart Meters".
- Medical device R&D systems received calibration and safety certification from Cairo University Medical Calibration and Safety Center; Egyptian Drug Authority licensing procedures are currently in progress.
- Public wording for sensitive communication work is intentionally limited to mission-critical communication platforms and secure communication systems.

Contact & Collaboration

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Open to senior technical leadership, consulting, partnerships, technical advisory, embedded/IoT product development, R&D strategy, and technology collaboration opportunities across Egypt, GCC, and remote global markets.